

Exploring Human-Generative AI Collaboration in Software Testing

Informed Consent

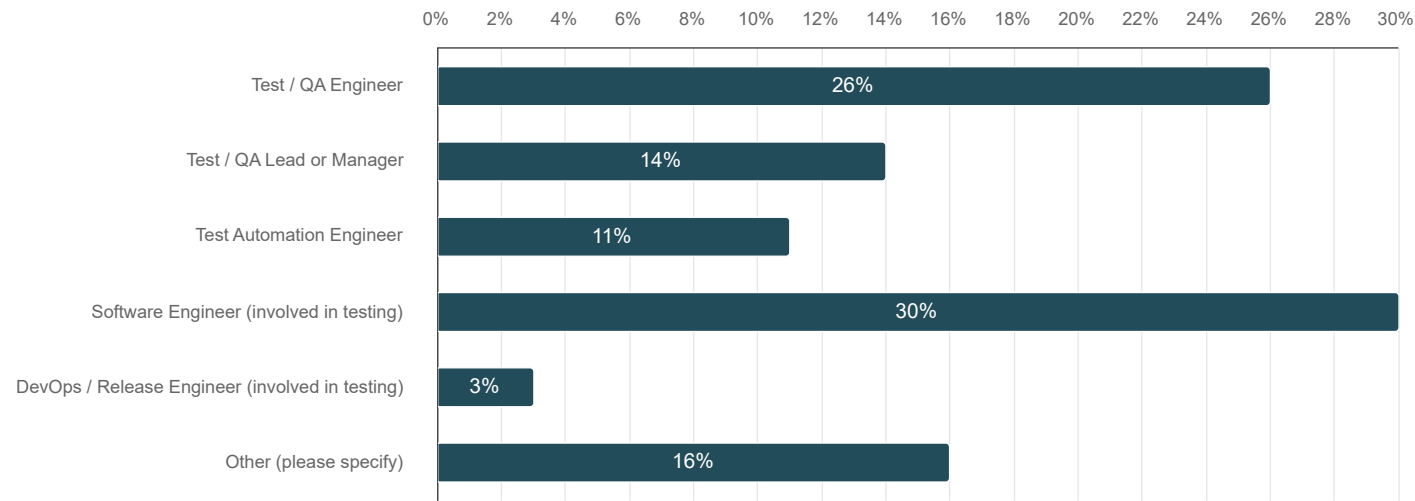
0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

Category	Percentage
All	100%

	n	Percent
I confirm that I have read and understood the Informed Consent for Participation in Research and voluntarily agree to participate in this research study.	64	100.0%

What is your current (or most recent) role related to software testing or quality assurance? (Select the closest match.)

Number of respondents: 64



	n	Percent
Test / QA Engineer	17	26.6%
Test / QA Lead or Manager	9	14.1%
Test Automation Engineer	7	10.9%
Software Engineer (involved in testing)	19	29.7%
DevOps / Release Engineer (involved in testing)	2	3.1%
Other (please specify)	10	15.6%

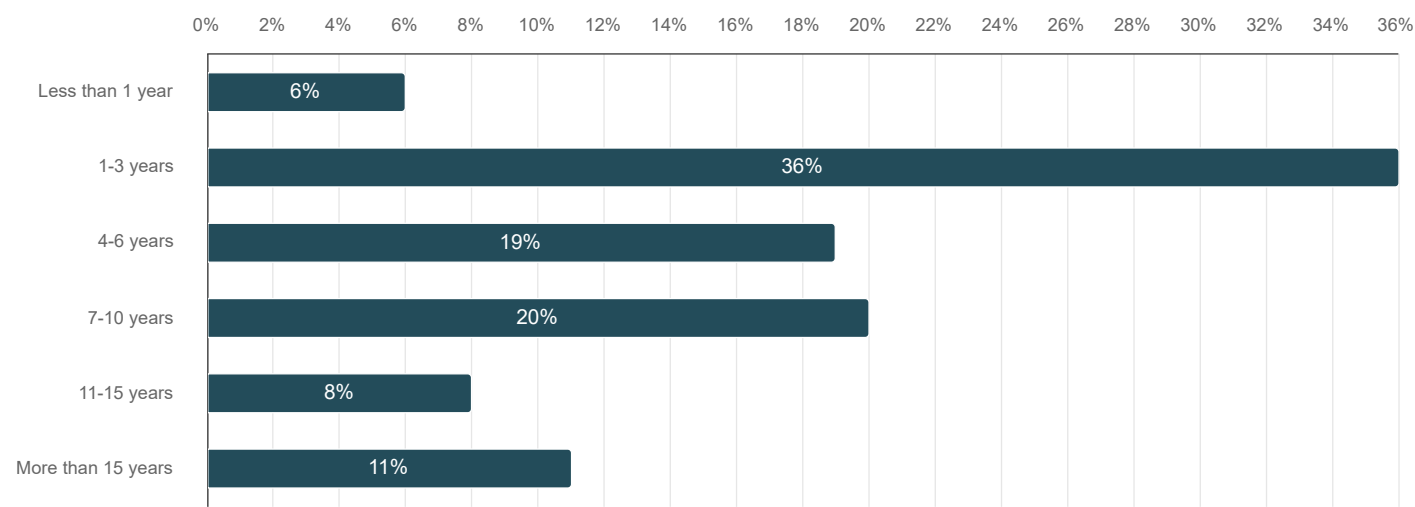
Answers given into textfield

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Option names	Text
Other (please specify)	Product owner
Other (please specify)	Product Validation
Other (please specify)	business analyst
Other (please specify)	Business Process Analyst involved in Testing as well
Other (please specify)	IT instructor
Other (please specify)	Junior Researcher
Other (please specify)	Machine Learning Engineer
Other (please specify)	Project Manager
Other (please specify)	Software engineer with expertise in DevOps and experience with some testing frameworks
Other (please specify)	Research Engineer

How many years of experience do you have in software testing or QA?

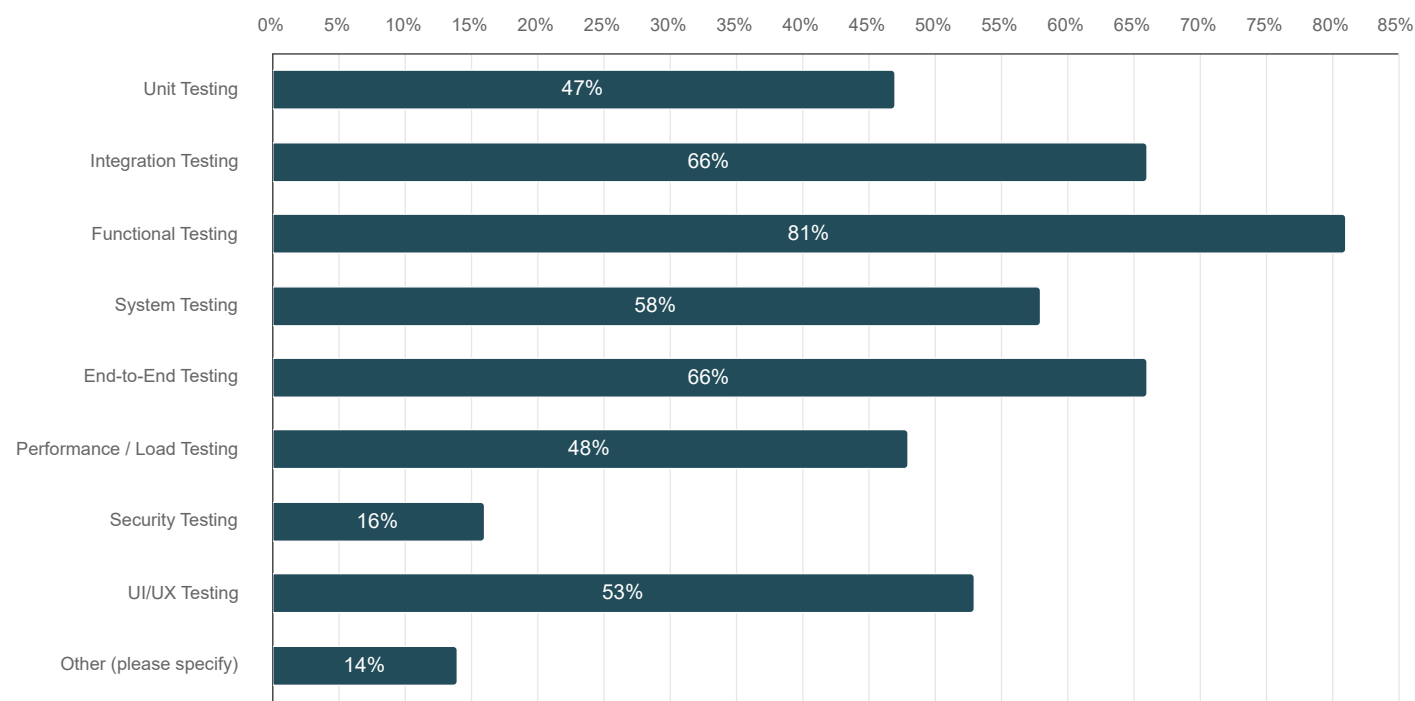
Number of respondents: 64



	n	Percent
Less than 1 year	4	6.3%
1-3 years	23	35.9%
4-6 years	12	18.8%
7-10 years	13	20.3%
11-15 years	5	7.8%
More than 15 years	7	10.9%

Which of the following types of testing do you regularly engage in? (Select all that apply)

Number of respondents: 64 , selected answers: 287



	n	Percent
Unit Testing	30	46.9%
Integration Testing	42	65.6%
Functional Testing	52	81.3%
System Testing	37	57.8%
End-to-End Testing	42	65.6%
Performance / Load Testing	31	48.4%
Security Testing	10	15.6%
UI/UX Testing	34	53.1%
Other (please specify)	9	14.1%

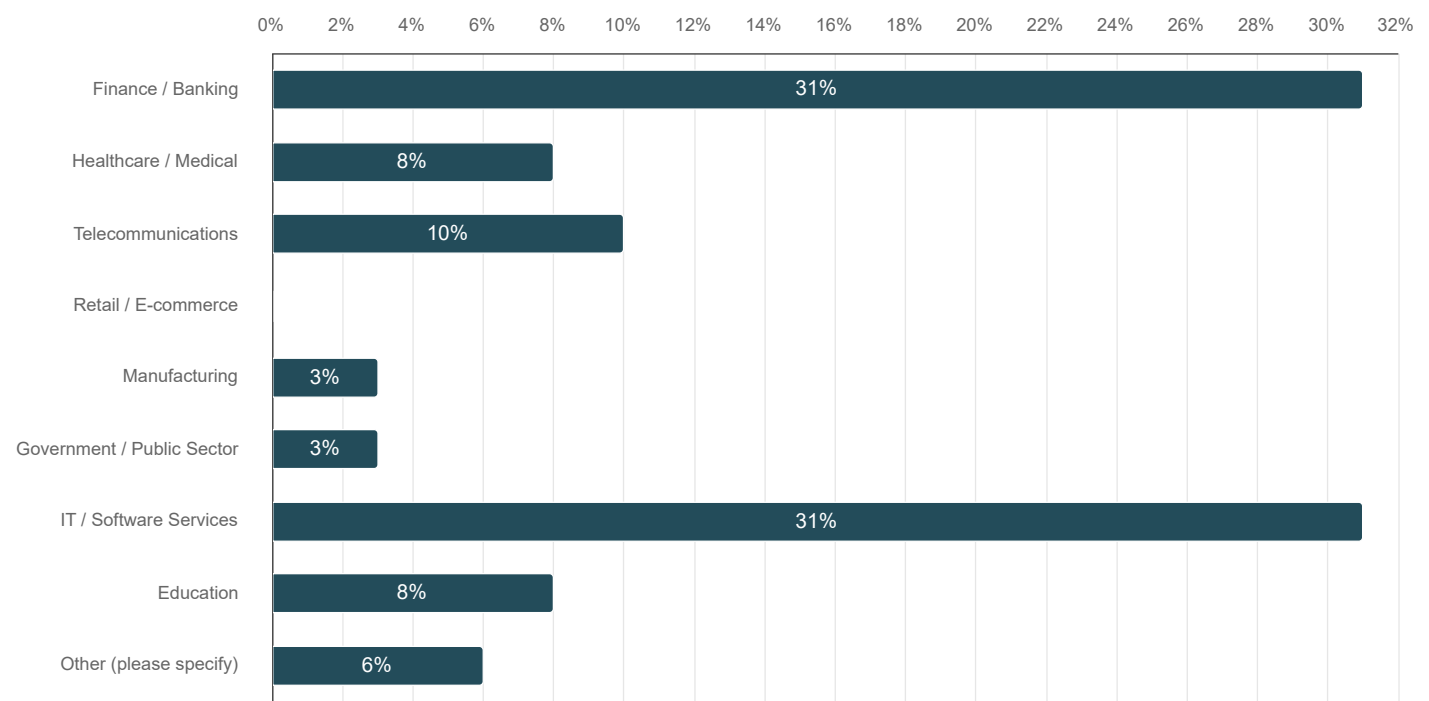
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Option names	Text
Other (please specify)	Test tools and environment development
Other (please specify)	UAT
Other (please specify)	Adhoc testing, Regression Testing
Other (please specify)	Regression Testing
Other (please specify)	Exploratory
Other (please specify)	API
Other (please specify)	smoke testing - ensuring basics are working before starting functional testing
Other (please specify)	API testing
Other (please specify)	API Testing

In which business domain or industry does your company primarily operate? (Select the closest match)

Number of respondents: 64



	n	Percent
Finance / Banking	20	31.2%
Healthcare / Medical	5	7.8%
Telecommunications	6	9.4%
Retail / E-commerce	0	0.0%
Manufacturing	2	3.1%
Government / Public Sector	2	3.1%
IT / Software Services	20	31.3%
Education	5	7.8%
Other (please specify)	4	6.3%

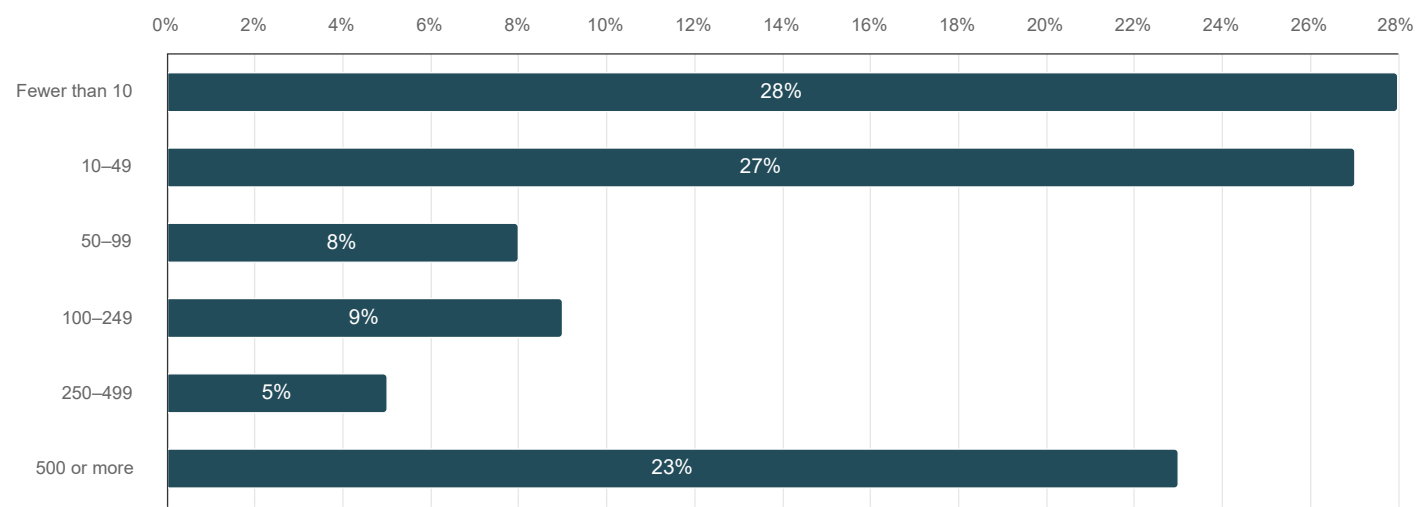
Answers given into textfield

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Option names	Text
Other (please specify)	Marine
Other (please specify)	Gambling
Other (please specify)	Travel insurance
Other (please specify)	Automotive infotainment

Approximately how many people are involved in software development (including testing) in your organization?

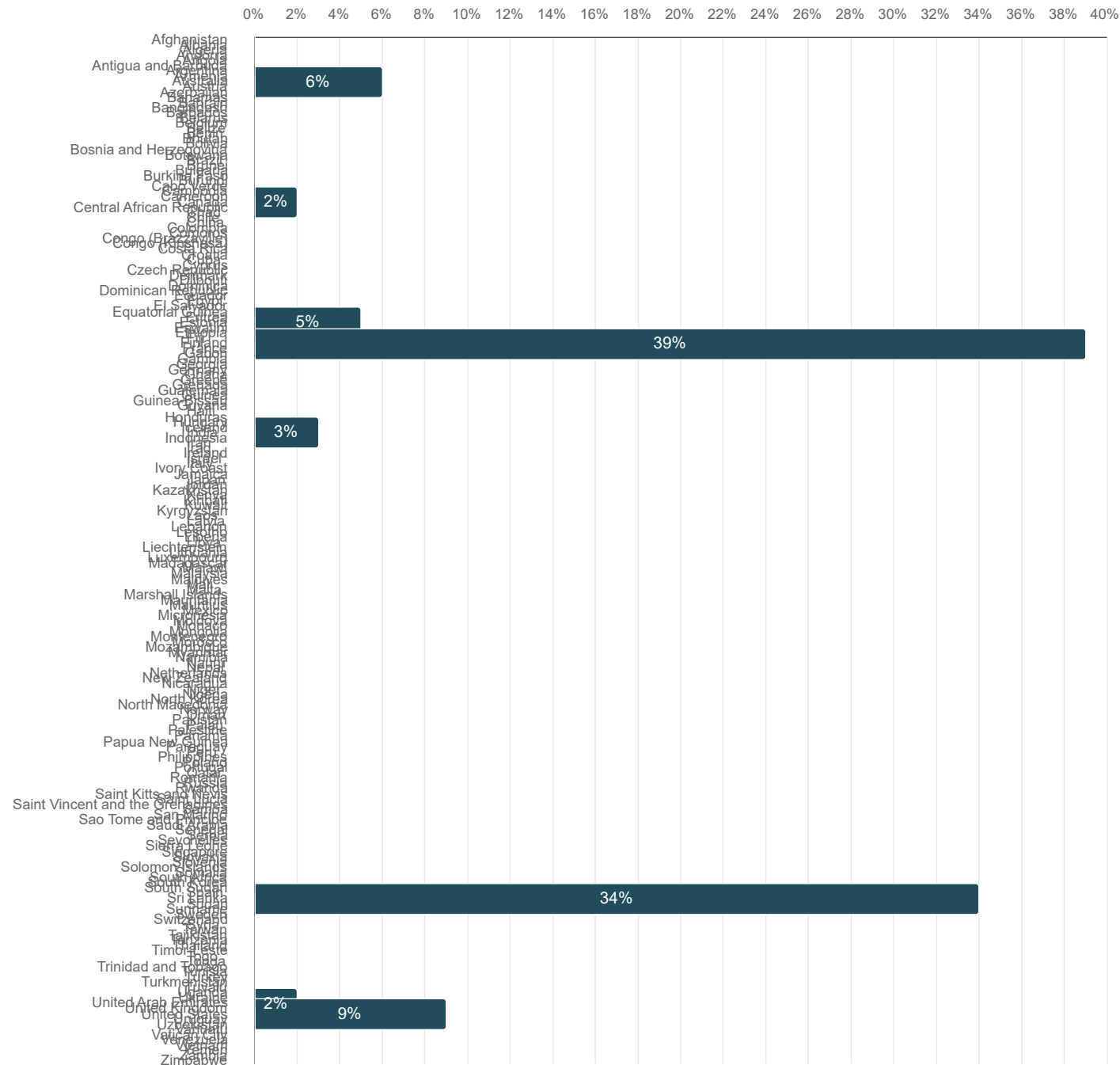
Number of respondents: 64



	n	Percent
Fewer than 10	18	28.1%
10-49	17	26.6%
50-99	5	7.8%
100-249	6	9.4%
250-499	3	4.7%
500 or more	15	23.4%

In which country are you currently based?

Number of respondents: 64



	n	Percent
Afghanistan	0	0.0%
Albania	0	0.0%
Algeria	0	0.0%
Andorra	0	0.0%
Angola	0	0.0%
Antigua and Barbuda	0	0.0%
Argentina	0	0.0%
Armenia	0	0.0%
Australia	4	6.2%
Austria	0	0.0%
Azerbaijan	0	0.0%
Bahamas	0	0.0%

	n	Percent
Bahrain	0	0.0%
Bangladesh	0	0.0%
Barbados	0	0.0%
Belarus	0	0.0%
Belgium	0	0.0%
Belize	0	0.0%
Benin	0	0.0%
Bhutan	0	0.0%
Bolivia	0	0.0%
Bosnia and Herzegovina	0	0.0%
Botswana	0	0.0%
Brazil	0	0.0%
Brunei	0	0.0%
Bulgaria	0	0.0%
Burkina Faso	0	0.0%
Burundi	0	0.0%
Cabo Verde	0	0.0%
Cambodia	0	0.0%
Cameroon	0	0.0%
Canada	1	1.6%
Central African Republic	0	0.0%
Chad	0	0.0%
Chile	0	0.0%
China	0	0.0%
Colombia	0	0.0%
Comoros	0	0.0%
Congo (Brazzaville)	0	0.0%
Congo (Kinshasa)	0	0.0%
Costa Rica	0	0.0%
Croatia	0	0.0%
Cuba	0	0.0%
Cyprus	0	0.0%
Czech Republic	0	0.0%
Denmark	0	0.0%
Djibouti	0	0.0%
Dominica	0	0.0%
Dominican Republic	0	0.0%
Ecuador	0	0.0%
Egypt	0	0.0%
El Salvador	0	0.0%
Equatorial Guinea	0	0.0%
Eritrea	0	0.0%
Estonia	3	4.7%

	n	Percent
Eswatini	0	0.0%
Ethiopia	0	0.0%
Fiji	0	0.0%
Finland	25	39.0%
France	0	0.0%
Gabon	0	0.0%
Gambia	0	0.0%
Georgia	0	0.0%
Germany	0	0.0%
Ghana	0	0.0%
Greece	0	0.0%
Grenada	0	0.0%
Guatemala	0	0.0%
Guinea	0	0.0%
Guinea-Bissau	0	0.0%
Guyana	0	0.0%
Haiti	0	0.0%
Honduras	0	0.0%
Hungary	0	0.0%
Iceland	0	0.0%
India	2	3.1%
Indonesia	0	0.0%
Iran	0	0.0%
Iraq	0	0.0%
Ireland	0	0.0%
Israel	0	0.0%
Italy	0	0.0%
Ivory Coast	0	0.0%
Jamaica	0	0.0%
Japan	0	0.0%
Jordan	0	0.0%
Kazakhstan	0	0.0%
Kenya	0	0.0%
Kiribati	0	0.0%
Kuwait	0	0.0%
Kyrgyzstan	0	0.0%
Laos	0	0.0%
Latvia	0	0.0%
Lebanon	0	0.0%
Lesotho	0	0.0%
Liberia	0	0.0%
Libya	0	0.0%
Liechtenstein	0	0.0%

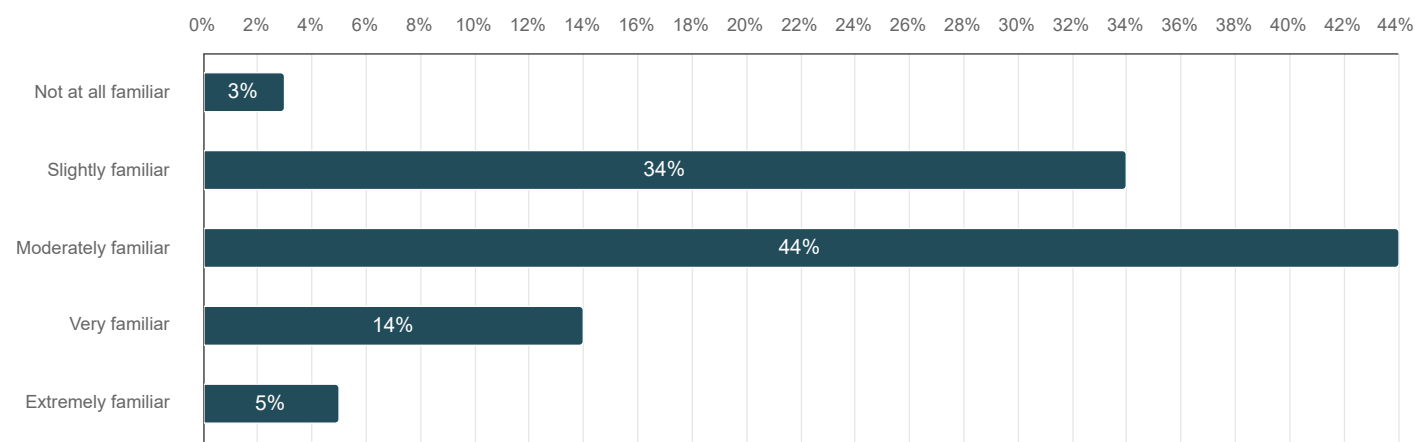
	n	Percent
Lithuania	0	0.0%
Luxembourg	0	0.0%
Madagascar	0	0.0%
Malawi	0	0.0%
Malaysia	0	0.0%
Maldives	0	0.0%
Mali	0	0.0%
Malta	0	0.0%
Marshall Islands	0	0.0%
Mauritania	0	0.0%
Mauritius	0	0.0%
Mexico	0	0.0%
Micronesia	0	0.0%
Moldova	0	0.0%
Monaco	0	0.0%
Mongolia	0	0.0%
Montenegro	0	0.0%
Morocco	0	0.0%
Mozambique	0	0.0%
Myanmar	0	0.0%
Namibia	0	0.0%
Nauru	0	0.0%
Nepal	0	0.0%
Netherlands	0	0.0%
New Zealand	0	0.0%
Nicaragua	0	0.0%
Niger	0	0.0%
Nigeria	0	0.0%
North Korea	0	0.0%
North Macedonia	0	0.0%
Norway	0	0.0%
Oman	0	0.0%
Pakistan	0	0.0%
Palau	0	0.0%
Palestine	0	0.0%
Panama	0	0.0%
Papua New Guinea	0	0.0%
Paraguay	0	0.0%
Peru	0	0.0%
Philippines	0	0.0%
Poland	0	0.0%
Portugal	0	0.0%
Qatar	0	0.0%

	n	Percent
Romania	0	0.0%
Russia	0	0.0%
Rwanda	0	0.0%
Saint Kitts and Nevis	0	0.0%
Saint Lucia	0	0.0%
Saint Vincent and the Grenadines	0	0.0%
Samoa	0	0.0%
San Marino	0	0.0%
Sao Tome and Principe	0	0.0%
Saudi Arabia	0	0.0%
Senegal	0	0.0%
Serbia	0	0.0%
Seychelles	0	0.0%
Sierra Leone	0	0.0%
Singapore	0	0.0%
Slovakia	0	0.0%
Slovenia	0	0.0%
Solomon Islands	0	0.0%
Somalia	0	0.0%
South Africa	0	0.0%
South Korea	0	0.0%
South Sudan	0	0.0%
Spain	0	0.0%
Sri Lanka	22	34.4%
Sudan	0	0.0%
Suriname	0	0.0%
Sweden	0	0.0%
Switzerland	0	0.0%
Syria	0	0.0%
Taiwan	0	0.0%
Tajikistan	0	0.0%
Tanzania	0	0.0%
Thailand	0	0.0%
Timor-Leste	0	0.0%
Togo	0	0.0%
Tonga	0	0.0%
Trinidad and Tobago	0	0.0%
Tunisia	0	0.0%
Turkey	0	0.0%
Turkmenistan	0	0.0%
Tuvalu	0	0.0%
Uganda	0	0.0%
Ukraine	0	0.0%

	n	Percent
United Arab Emirates	1	1.6%
United Kingdom	0	0.0%
United States	6	9.4%
Uruguay	0	0.0%
Uzbekistan	0	0.0%
Vanuatu	0	0.0%
Vatican City	0	0.0%
Venezuela	0	0.0%
Vietnam	0	0.0%
Yemen	0	0.0%
Zambia	0	0.0%
Zimbabwe	0	0.0%

How familiar are you with Generative AI tools?

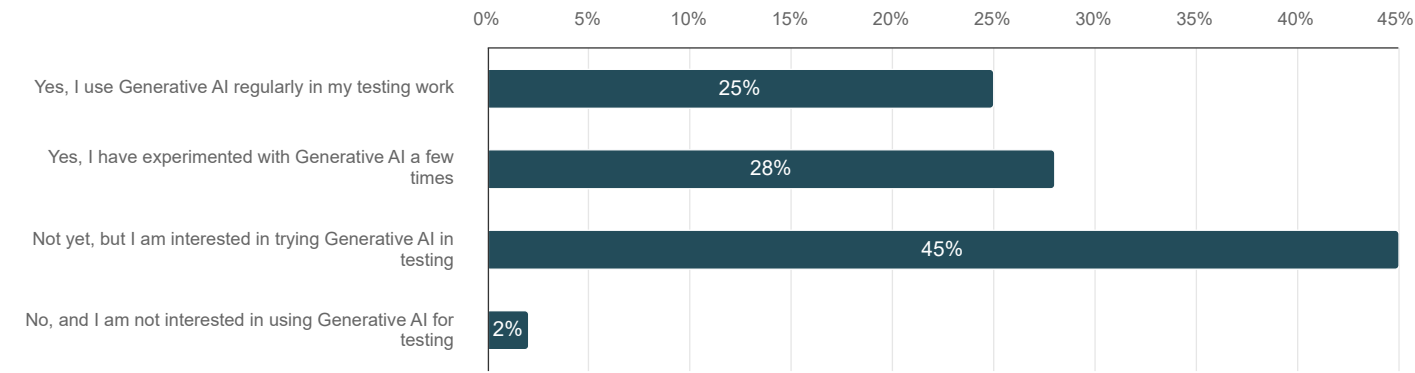
Number of respondents: 64



	n	Percent
Not at all familiar	2	3.1%
Slightly familiar	22	34.4%
Moderately familiar	28	43.7%
Very familiar	9	14.1%
Extremely familiar	3	4.7%

Have you personally used any Generative AI tools as part of your software testing work?

Number of respondents: 64



	n	Percent
Yes, I use Generative AI regularly in my testing work	16	25.0%
Yes, I have experimented with Generative AI a few times	18	28.1%
Not yet, but I am interested in trying Generative AI in testing	29	45.3%
No, and I am not interested in using Generative AI for testing	1	1.6%

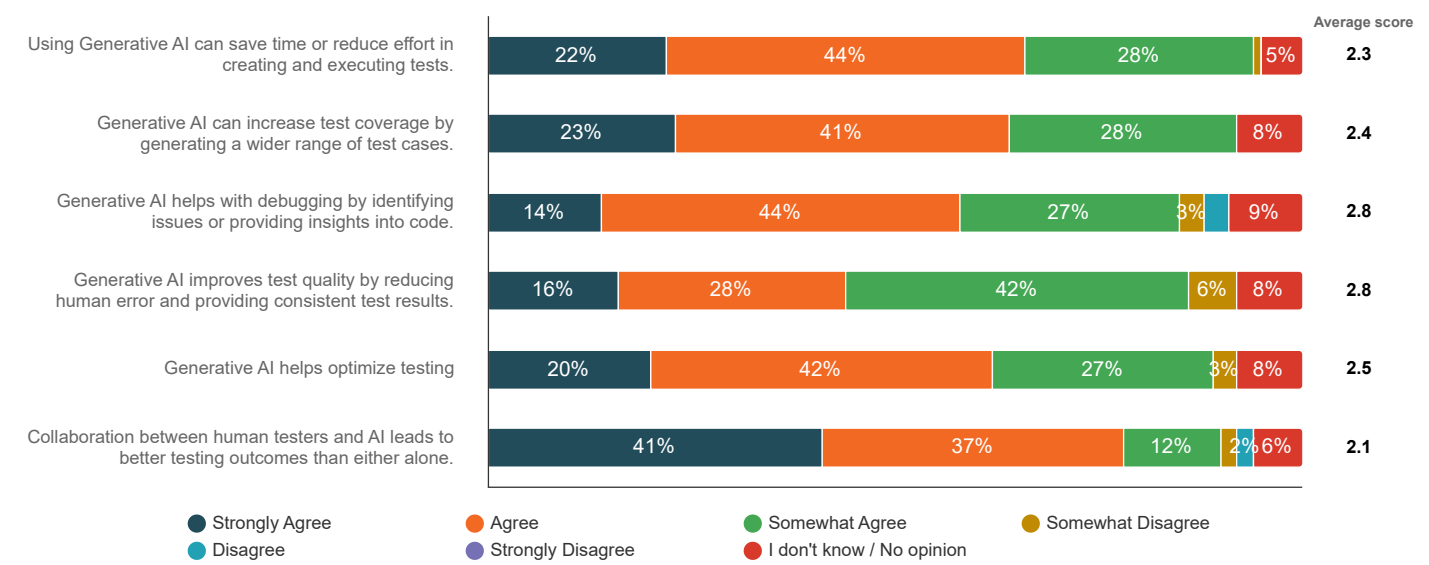
If you have used Generative AI in your testing work, please briefly describe one example of how you used it or what you used it for. (If you have not used Generative AI in testing, you may skip this question.)

Number of respondents: 25

Responses
I use generative ai tools for Creating test cases. Generating random test data. Generate validation rules in excel.
To support coding and test automation development.
I used testRigor AI tool to create automation test scripts.
For building solutions using AI which in turn helps Test Automation
For scripting Unit testcase idea generation
Test scope identification Scenario generation Troubleshooting Complex function scripting Code formatting and standardization.
- speeds up the code writing significantly - can handle UI testing nowadays - comes up with really good testing scenarios
basically for API testing to test actual output with expected output in different scenarios
If tests are failing, we can use an AI tool to identify the possible root cause and determine what should be checked first
- make a test requirement template use it to create test requirements - write test cases - write unit tests, integration tests -analyse test fail with copilot
While doing automation it helps me to find the error and sort it out very quickly
To come up with test scenarios To generate functions to do a certain task Asking coding questions
As a BA I support QAs with test planning. I use Gen AI for test planning.
Test case generation test data creation and make test case bots as well such as unit test bots E2E test bots
I have explicitly asked the models to write unit test for each and every features and make sure the line coverage is above expected.
NA
Asking examples about good test cases for cucumber scenarios related to API test automation. Also asking help if needed for developing the code behind cucumber steps (Helper files). Also asked help for unit testing like how to serialize tests etc and what kind of unit tests needed based on source code.
Test case generation
Generate automation code
Copilot
My team has used it to create unit testing for a new software we created
Generate test cases, create Jira bug tickets
We have a locally integrated GPT within our application that explains the functionality of each page and generates real-time messages for any threats or anomalies detected for our customers. I need to test the GPT to ensure it provides clear, customer-friendly explanations and generates accurate, relevant messages.
-
In debugging a new problem: as modern LLMs can have big codebases as context, I usually start by asking the AI where I can find relevant code for the issue, then ask what could be the cause of the issue in the relevant section of code, then ask it to implement a fix and personally verify results.

Please indicate how much you agree or disagree with each of the following statements about the potential benefits of using Generative AI in software testing.

Number of respondents: 64



	Strongly Agree	Agree	Somewhat Agree	Somewhat Disagree	Disagree	Strongly Disagree	I don't know / No opinion	Average	Median
Using Generative AI can save time or reduce effort in creating and executing tests.	21.9%	43.7%	28.1%	1.6%	0.0%	0.0%	4.7%	2.3	2.0
Generative AI can increase test coverage by generating a wider range of test cases.	23.5%	40.6%	28.1%	0.0%	0.0%	0.0%	7.8%	2.4	2.0
Generative AI helps with debugging by identifying issues or providing insights into code.	14.1%	43.7%	26.6%	3.1%	3.1%	0.0%	9.4%	2.8	2.0
Generative AI improves test quality by reducing human error and providing consistent test results.	15.6%	28.1%	42.2%	6.3%	0.0%	0.0%	7.8%	2.8	3.0
Generative AI helps optimize testing	20.3%	42.2%	26.6%	3.1%	0.0%	0.0%	7.8%	2.5	2.0
Collaboration between human testers and AI leads to better testing outcomes than either alone.	40.6%	37.5%	12.5%	1.6%	1.6%	0.0%	6.2%	2.1	2.0
Total	22.7%	39.3%	27.4%	2.6%	0.8%	0.0%	7.3%	2.5	2.0

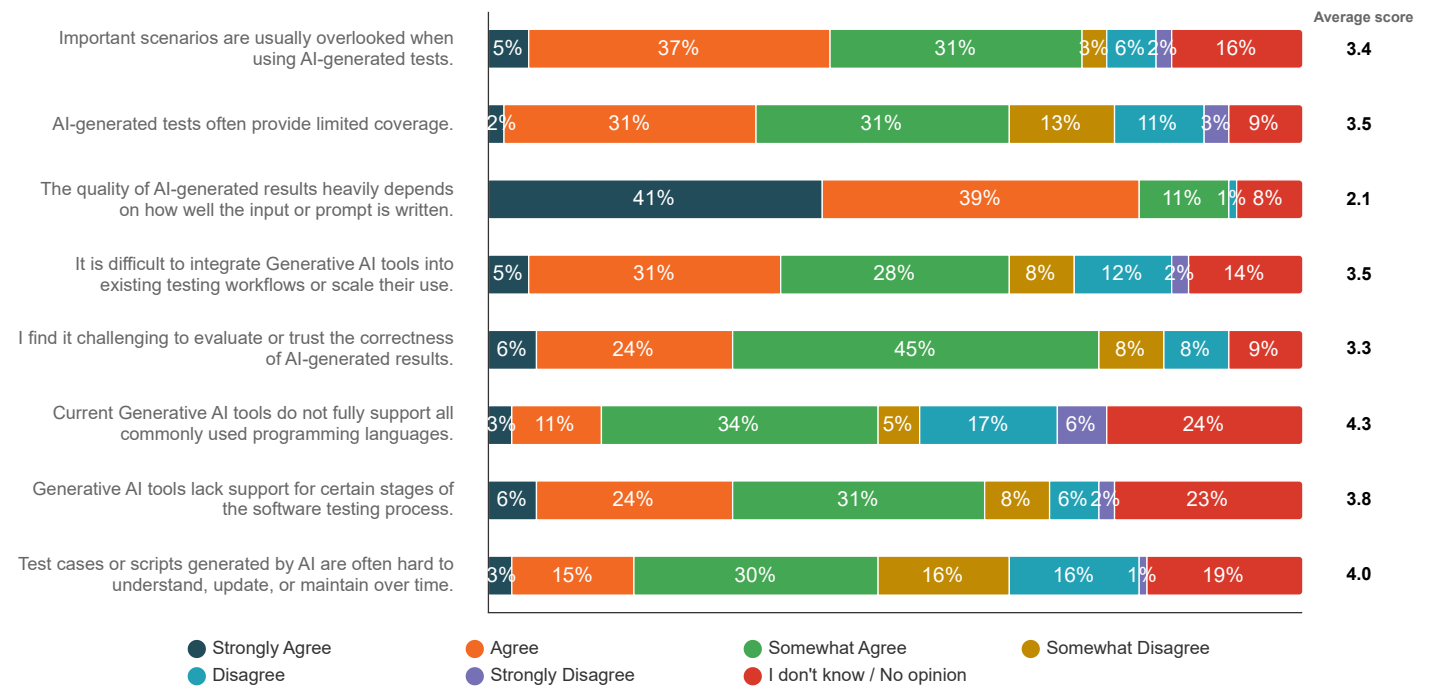
Please indicate any other benefits of using Generative AI in testing.

Number of respondents: 13

Responses
Quicker test case creation and gives ideas how to do testing. At this time I would not only rely on AI testing only. Maybe in the future it is possible. Test data creating on different purposes in for example banking services is very challenging process. AI could be helpful with that when creating some mocked test data for example with performance testing needs when you need large amounts of data.
Helps to generate test cases
For commenting and generate descriptions of bugs and fixes
Quick troubleshooting. Element identification is very quick (Xpaths , Ids)
Time saving Test scenario identification Root cause identification
Generative Ai works well when we use the right prompt. prompt is the key to productive code
Fast delivery and best quality
Test data preparation Test status reporting
It helps to save time and not writing all code by myself. Also it gives hints about good test cases.
Adopting to the new techs for daily work
It is useful in generating edge cases for the testing
Automate test case generation, speed up test execution, identify edge cases, and enhance test coverage efficiently.
Comeup with unanticipated scenarios that could happen in realtime production data using AI tools

Please indicate your level of agreement with each of the following statements about challenges in using Generative AI for software testing.

Number of respondents: 64



	Strongly Agree	Agree	Somewhat Agree	Somewhat Disagree	Disagree	Strongly Disagree	I don't know / No opinion	Average	Median
Important scenarios are usually overlooked when using AI-generated tests.	4.7%	37.5%	31.2%	3.1%	6.3%	1.6%	15.6%	3.4	3.0
AI-generated tests often provide limited coverage.	1.6%	31.2%	31.3%	12.5%	10.9%	3.1%	9.4%	3.5	3.0
The quality of AI-generated results heavily depends on how well the input or prompt is written.	40.6%	39.1%	10.9%	0.0%	1.6%	0.0%	7.8%	2.1	2.0
It is difficult to integrate Generative AI tools into existing testing workflows or scale their use.	4.7%	31.2%	28.1%	7.8%	12.5%	1.6%	14.1%	3.5	3.0
I find it challenging to evaluate or trust the correctness of AI-generated results.	6.3%	23.4%	45.3%	7.8%	7.8%	0.0%	9.4%	3.3	3.0
Current Generative AI tools do not fully support all commonly used programming languages.	3.1%	10.9%	34.4%	4.7%	17.2%	6.3%	23.4%	4.3	4.0
Generative AI tools lack support for certain stages of the software testing process.	6.3%	23.4%	31.2%	7.8%	6.3%	1.6%	23.4%	3.8	3.0
Test cases or scripts generated by AI are often hard to understand, update, or maintain over time.	3.1%	15.6%	29.7%	15.6%	15.6%	1.6%	18.8%	4.0	4.0
Total	8.8%	26.5%	30.3%	7.4%	9.8%	2.0%	15.2%	3.5	3.0

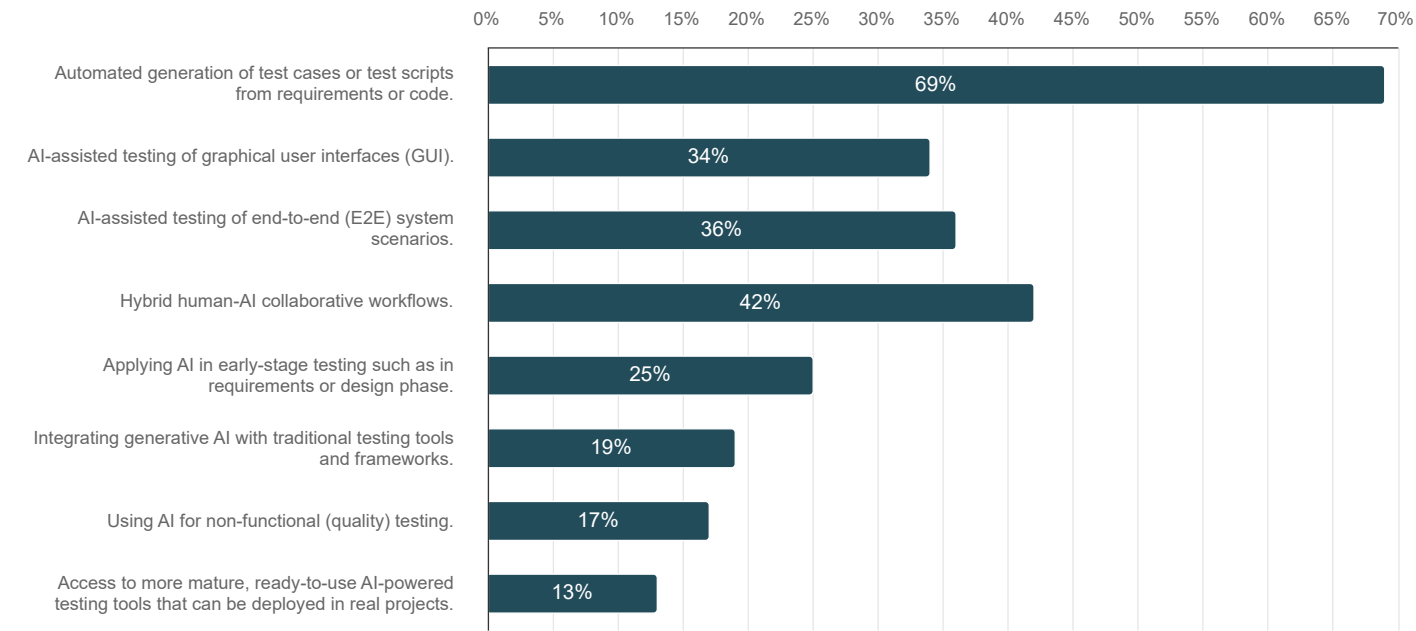
Please indicate any other challenges of using Generative AI in software testing.

Number of respondents: 6

Responses
I would add one test a time when creating tests. I don't like to have hundreds of tests testing something. I like to have small efficient easy to maintain test set that can be run daily basis. It will let me know if something goes broken instantly.
Difficult to make it understand the scenario which are specific to organisation
Sometimes AI tools are not compatible with complex UI functionalities.
Because of company restrictions, sometimes it is not possible to put all information into an AI tool
It's not covering the regression and adhoc scenarios
Lack of transparency in how decisions or test cases are generated, making debugging and validation difficult.

Which of the following potential applications of Generative AI in software testing do you find most promising or exciting? (Select up to 3)

Number of respondents: 64 , selected answers: 163



	n	Percent
Automated generation of test cases or test scripts from requirements or code.	44	68.8%
AI-assisted testing of graphical user interfaces (GUI).	22	34.4%
AI-assisted testing of end-to-end (E2E) system scenarios.	23	35.9%
Hybrid human-AI collaborative workflows.	27	42.2%
Applying AI in early-stage testing such as in requirements or design phase.	16	25.0%
Integrating generative AI with traditional testing tools and frameworks.	12	18.8%
Using AI for non-functional (quality) testing.	11	17.2%
Access to more mature, ready-to-use AI-powered testing tools that can be deployed in real projects.	8	12.5%

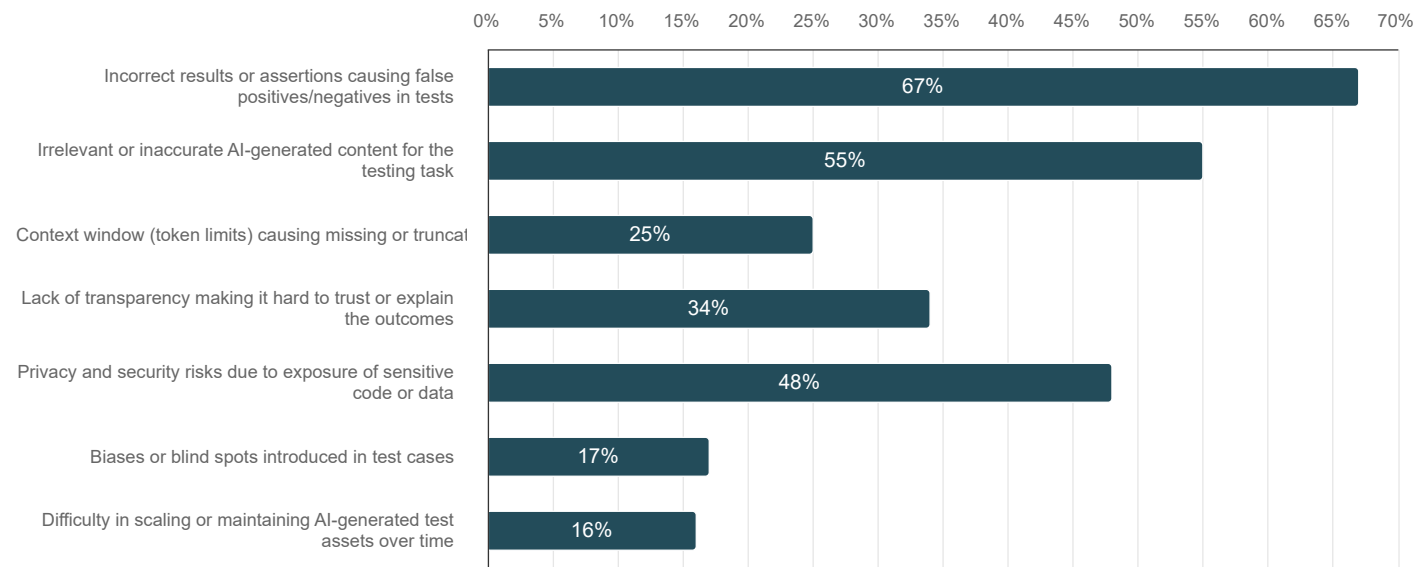
Please indicate any other promising opportunities of using Generative AI in software testing.

Number of respondents: 6

Responses
For example take a screen shot of tested system and tell ChatGPT to generate test automation for these. No need to write everything by hand. Saves tons of work and time.
I would say literally all the options mentioned at point 15
Test case can be automated
Proactive system monitoring and identifying risk areas for testing based on changes to the code base
Code reviews, automated tests generation
Gen AI can help accelerate regression testing by quickly adapting test cases to code changes

Which of the following risks or concerns about using Generative AI in software testing do you consider most critical? (Select up to 3)

Number of respondents: 64 , selected answers: 168



	n	Percent
Incorrect results or assertions causing false positives/negatives in tests	43	67.2%
Irrelevant or inaccurate AI-generated content for the testing task	35	54.7%
Context window (token limits) causing missing or truncated information, leading to incomplete analysis or suggestions	16	25.0%
Lack of transparency making it hard to trust or explain the outcomes	22	34.4%
Privacy and security risks due to exposure of sensitive code or data	31	48.4%
Biases or blind spots introduced in test cases	11	17.2%
Difficulty in scaling or maintaining AI-generated test assets over time	10	15.6%

Please indicate any other risks or concerns of using Generative AI in software testing.

Number of respondents: 3

Responses
With testing I prefer more "less high quality test cases" is better than "hundreds of bad or irrelevant test cases". I would rather select few hand picked tests for critical areas of the functionality, than run all AI generated sometimes irrelevant tests. It is important that when test run results are being checked, everybody or at least the tester itself understands what is going on with the service just by looking the test results. I believe AI testing could give false positive or false negative results easily. In time these kind of test runs goes irrelevant and nobody sees reason to looking after them on daily basis. The whole service is good to understand and try out different interfaces and integration points with few tests especially when executing system level testing for any service.
I think the major risk might be security: - information that is fed to the AI - and "AI" tools that are created by bad actors to get your data
Due to company privacy policies, it is not possible to put all the data to the AI tool

Do you have any other comments, suggestions, or experiences regarding human-AI collaboration in software testing that you would like to share?

Number of respondents: 17

Responses
I think AI is still taking first baby steps on testing or then I just don't know enough of this subject myself. Some people may use it more than me, but I cant wait to see next 10 years what comes up and how this field will change the ways of working.
When the tasks/the system is complex the AI often fails to understand and perform accurate tests.
I never tried this. So I don't have any comments
AI is great for brainstorming and idea generation. I think we can integrate some parts of the software design flow with Ai. But human intervention is mandatory.
AI tools are useful since it is still developing we have to double-check the results
N/A
AI is good for improving test coverage and requirement matching with respect to requirement documents, but it is hard to address the user experience part of testing.
If an AI tool can analyze test results when they fail and show where the failure occurred, it will make analysis much easier
As I have read, most MCP servers using AI helps generate more accurate results as we can provide access using the MCP protocol to the given tool to communicate with the code base and provide more secured and accurate results
Generally testers add value as the first point of triage of the defects. generative AI can help with prioritising defects.
People should learn to prompt properly to get the benefits
Gen AI will provide a great learning curve for a novice tester in the future
Human understanding of the domain of test g can go lower due to excessive usage of AI
I feel like software testing is a field that can have a huge impact from current AI trends. We are actively using it to generate test cases and generate synthetic data for our applications. It helps us to not to expose our customers data for testing
In my experience, combining AI-generated suggestions with testers leads to more reliable and efficient outcomes. And also communication and feedback loops between testers and AI tools are more essential.
Human-AI collaboration in software testing is evolving from simple automation to creative partnership. Testers can now use AI to generate edge cases, simulate diverse user behaviors, and uncover patterns in defects. Instead of replacing testers, AI becomes a thinking partner enhancing exploration, strategy, and speed without losing human insight.
Like with any other use cases for AI, in testing one must, absolutely must consider the human element and not blindly jump into the hands of AI.